

Day 1 — The Hook: $2_3 = 253$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 2-digit $\times 11$ product where the digit-sum is less than 10, in under 3 seconds, and will have recorded a personal baseline time for the same problem using standard long multiplication.

Materials

- Printed baseline sheet: ONE multiplication problem (73×11). One per student.
- Stopwatch or wall clock with a second hand. (One per pair is plenty.)
- Whiteboard / large paper to demonstrate.

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: ekādhikena pūrveṇa; lit. “by one more than the previous”) — the Sanskrit name Tirthaji (1965) gives the underlying sūtra. We use it as the cultural anchor; the technique is what students take home.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to compute 73×11 the normal way? Make a guess.”
- Take 3–4 guesses. Write the range on the board.

Baseline timing (8 min)

- Hand out the baseline sheet face-down. Students write their name and predicted time on the back.
- On “Go,” flip and solve 73×11 using long multiplication.
- Record actual time honestly. Mark answer themselves.
- Expected range: 15–60 seconds.

Teacher demonstration (12 min) — The trick

- Write on the board:

$$73 \times 11 = ?$$

- “I’m going to show you a method that takes about 2 seconds, then we’ll figure out why it works.”

Step 1: Write the digits, with a gap: 7_3

Step 2: Add the two digits: $7 + 3 = 10$.

Wait – that's the carry case. Park it.

- *Reset. Different number first. Try 23×11 :*

Step 1: Split: 2 _ 3
Step 2: Add: 2 + 3 = 5
Step 3: Drop in: 2 5 3 → 253.

Check on the board with long multiplication: $23 \times 11 = 23 \times 10 + 23 = 230 + 23 = 253$. ✓

- Try 45×11 :

4 _ 5 → 4 + 5 = 9 → 4 9 5 → 495.

- Try 62×11 :

6 _ 2 → 6 + 2 = 8 → 6 8 2 → 682.

- “Pattern: outer digits stay the same; middle digit = their sum.”
- Now name it. “*Tirthaji (1965) calls this **Ekādhikena Pūrveṇa** — ‘by one more than the previous.’ That is the Sanskrit name of the sūtra. The trick itself is just $(10a + b) \times 11 = 100a + 10(a+b) + b$, which we’ll prove on Day 3.*”

Activity (15 min) — Try it yourself

- Worksheet (handed out): 10 two-digit $\times 11$ problems, ALL with digit-sum < 10 .

12×11	23×11	34×11	45×11	52×11
61×11	16×11	27×11	35×11	44×11

- Pair work. Partner times each other. Goal: each problem under 3 seconds.
- After 10 min, students re-do 73×11 on the back of the baseline sheet. *Hmm — $7 + 3 = 10$. That’s tomorrow’s lesson.*

Wrap-up (5 min)

- Show of hands: who finished all 10 in under 30 seconds? (Most will.)
- Tease tomorrow: “ 73×11 broke the rule because $7 + 3 = 10$ and you can’t write 10 in the middle of a number. Tomorrow — the carry case.”

Student-facing content

The $\times 11$ Trick (no-carry case)

For any 2-digit number with digit-sum < 10 : 1. **Split** the digits with a gap. 2. **Add** the two digits. 3. **Drop** the sum into the gap.

Example: $34 \times 11 \rightarrow 3 _ 4 \rightarrow 3+4 = 7 \rightarrow 374$.

Tirthaji (1965) names the underlying sūtra *Ekādhikena Pūrveṇa* — “by one more than the previous.” The trick is older than its modern Sanskrit name; the algebra (which we’ll do on Day 3) is the same in every language.

Homework

- Find any two 2-digit numbers (digit-sum < 10) in your house — phone numbers, page numbers, prices. Compute each $\times 11$ using the trick. Bring 5 worked examples to class tomorrow.

Differentiation

- **Tier up:** Try 3-digit $\times 11$ (e.g. 234×11 , 152×11). See if you can guess the pattern before Day 4.
- **Tier down:** If the gap is confusing, write the answer as a 3-digit number directly: hundreds = first digit, tens = sum, units = second digit.

Teacher notes

- **The baseline time matters.** Day 5’s retest depends on having a Day 1 record.
- Don’t dwell on the Sanskrit name today. *Ekādhikena Pūrveṇa* is the cultural anchor; the win is mental speed.
- Some students will already know the trick (especially those from competitive-math backgrounds). Have them be peer-coaches. Reframe their advantage as “you’ll learn the WHY this week.”
- About 30% of randomly-chosen 2-digit numbers have digit-sum ≥ 10 (the carry case). So today’s worksheet hand-picks the no-carry ones; tomorrow we add the carries.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. Sūtra 1, *Ekādhikena Pūrveṇa*.